

# EntraID (OIDC/JWT) Authentication for Claude Desktop

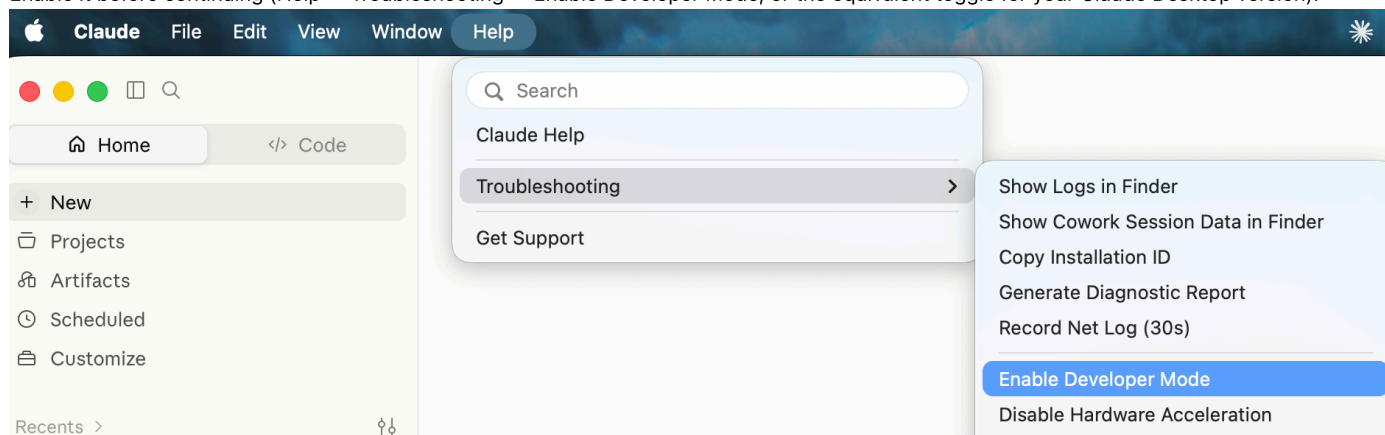
Configures Claude Desktop to authenticate against this self-hosted AIGW Gateway using Microsoft EntraID as the OIDC identity provider, with JWT bearer tokens validated by the gateway.

## Prerequisites

- This repo's `docker-compose.yml` stack running
  - `JWT_ENABLED: ON` (already set — see `docker-compose.yml`)
  - `JWT_LOCAL_AUTH_DEFAULT_SCOPES: completions.write,mcp.invoke` (already set — see `docker-compose.yml`)– This is required as EntraID is not setting the scope
- An EntraID tenant with permissions to register an application (entra.microsoft.com)
- Claude Desktop installed, with the Developer menu enabled (step 1 below)

## 1. Enable the Developer menu in Claude Desktop

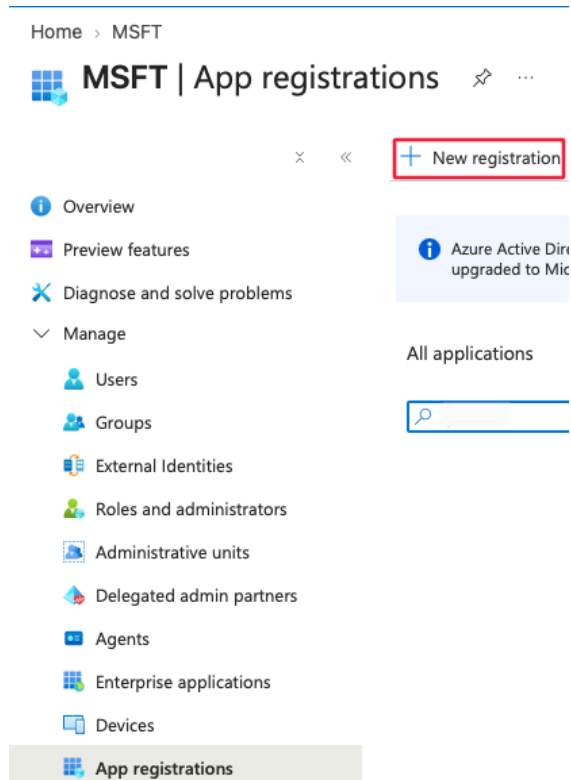
The 3rd-party provider / OIDC configuration UI referenced in step 6 lives behind Claude Desktop's Developer menu, which is hidden by default. Enable it before continuing (Help → Troubleshooting → Enable Developer Mode, or the equivalent toggle for your Claude Desktop version).



## 2. Register the application in EntraID

In the [Entra Admin Center](#):

1. Register a new application (App registrations → New registration).



2. Add a redirect URI for the **loopback port Claude Desktop will use** — see step 6 (<http://localhost:8080> in this example).

Home > MSFT | App registrations

Register an application

\* Name

The user-facing display name for this application (this can be changed later).

AIGW-OIDC

Supported account types

Choose the account types that can use this application or access this API

Single tenant only - MSFT

Help me choose

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Public client/native (mobile ...

http://127.0.0.1:8080/callback

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

3. Note the **Application (client) ID** and the **Application tenant ID** (You will use it as the **OIDC Issuer URL**<https://login.microsoftonline.com/<tenant-id>/v2.0>) — both are needed in step 6.

Home

AIGW-OIDC

Search

Deactivate Delete Endpoints

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Branding & properties

Authentication (Preview)

Certificate & secrets

Essentials

Display name : AIGW-OIDC

Application (client) ID : [redacted]

Object ID : [redacted]

Directory (tenant) ID : [redacted]

Supported account types : My organization only

Client credentials : Add a certificate or secret

Redirect URIs : 0 web, 0 spa, 1 public client

Application ID URI : Add an Application ID URI

Managed application in ... : AIGW-OIDC

State : Activated

4. **Set the token version to v2.** In the app **Manifest** tab, confirm "`accessTokenAcceptedVersion`" is 2, or patch it via Graph API:

```
PATCH https://graph.microsoft.com/v1.0/applications/{appObjectId}

Body: {"api": {"requestedAccessTokenVersion": 2}}
```

Without this, EntraID issues v1.0 tokens whose issuer (<https://sts.windows.net/<tenant-id>/>) won't match the v2.0 issuer URL configured in step 6, causing authentication to fail.

The `Setup-EntraID-App.sh` script applies this automatically.

3. Configure custom claims

The gateway requires specific claims in the ID token to identify the user and map them to the correct Portkey workspace. Several of these (e.g. `onPremisesSamAccountName`) are not emitted in a standard EntraID token, so they must be injected via an **EntraID Claims Mapping Policy** — a tenant-level policy that rewrites or adds claims before the token is issued.

**Tip:** The `Setup-EntraID-App.sh` script supplied with this deployment automates steps 3a–3d below, including creating the policy, assigning it to the service principal, and enabling `acceptMappedClaims`. You can use it instead of following the manual steps.

3a. Add optional ID-token claims

In the Entra Admin Center, go to **App registrations** → **your app** → **Token configuration** and add the following as optional claims on the **ID** token:

| Claim name         | Purpose              |
|--------------------|----------------------|
| <code>upn</code>   | User principal name  |
| <code>email</code> | User's email address |

| Claim name         | Purpose                   |
|--------------------|---------------------------|
| preferred_username | Display name / login hint |

These are available natively and can be added through the UI without a custom policy.

3b. Create a Claims Mapping Policy

The remaining claims require a Claims Mapping Policy, which must be created via the **Microsoft Graph API** or Azure CLI (there is no UI for this). The policy should be defined as follows, substituting your deployment-specific values where indicated:

```
{
  "definition": [
    {
      "ClaimsMappingPolicy": {
        "Version": 1,
        "IncludeBasicClaimSet": true,
        "ClaimsSchema": [
          {
            "Source": "user",
            "ID": "jobtitle",
            "JwtClaimType": "jobtitle"
          },
          {
            "Source": "user",
            "ID": "department",
            "JwtClaimType": "department"
          },
          {
            "Source": "user",
            "ID": "onpremisesamaccountname",
            "JwtClaimType": "uid"
          },
          {
            "Source": "user",
            "ID": "mailnickname",
            "JwtClaimType": "mailnickname"
          },
          {
            "Source": "user",
            "ID": "mail",
            "JwtClaimType": "email_id"
          },
          {
            "Source": "user",
            "ID": "mailnickname",
            "JwtClaimType": "_user"
          },
          {
            "Value": "<your-workspace-slug>",
            "JwtClaimType": "portkey_workspace"
          },
          {
            "Value": "<your-ORGANISATIONS_TO_SYNC-uuid>",
            "JwtClaimType": "portkey_oid"
          }
        ]
      }
    ]
  },
  "displayName": "PortkeyClaimsMappingPolicy",
  "type": "ClaimsMappingPolicy"
}
```

The claims this policy injects are:

| JWT claim         | Source                                     | Notes                                                     |
|-------------------|--------------------------------------------|-----------------------------------------------------------|
| jobtitle          | user.jobtitle                              |                                                           |
| department        | user.department                            |                                                           |
| uid               | user.onPremisesSamAccountName              | Requires on-prem AD sync                                  |
| mailnickname      | user.mailNickname                          |                                                           |
| email_id          | user.mail                                  | Primary user identity for the gateway                     |
| _user             | user.mailNickname                          | Internal user identifier                                  |
| portkey_workspace | Static value <b>or</b> extension attribute | See note below                                            |
| portkey_oid       | Static value                               | Must match your deployment's <b>ORGANISATIONS_TO_SYNC</b> |

The most load-bearing claims are **email\_id** (user identity) and **portkey\_oid** (org routing). The gateway will reject tokens that are missing either.

Sourcing **portkey\_workspace**

You have two options for this claim:

| Option                   | Policy entry                                                                                      | When to use                                                                                                                |
|--------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Static value             | <code>{"Value": "ws-main-a-997260", "JwtClaimType": "portkey_workspace"}</code>                   | All users share a single workspace                                                                                         |
| User extension attribute | <code>{"Source": "user", "ID": "extensionattribute1", "JwtClaimType": "portkey_workspace"}</code> | Per-user workspace assignment — set the workspace slug on each user's <b>extensionAttribute1-15</b> in Entra or on-prem AD |

The extension-attribute approach is recommended for multi-workspace deployments: the value is read from the user object at token issuance time, so users can be reassigned to different workspaces without any policy change.

**Note on group-based sourcing:** Claims Mapping Policies do not support deriving a single claim value from group membership — the `groups` claim returns an array of GUIDs, not a workspace slug. If you need workspace assignment driven by group membership, use [dynamic group rules](#) to keep membership in sync, and write the workspace slug back to an `extensionAttribute` on the user via a provisioning flow.

### 3c. Assign the policy to the service principal

After creating the policy, assign it to the app's service principal:

```
POST https://graph.microsoft.com/v1.0/servicePrincipals/{servicePrincipalId}/claimsMappingPolicies/$ref
Body: {"@odata.id": "https://graph.microsoft.com/v1.0/policies/claimsMappingPolicies/{policyId}"}
```

### 3d. Enable `acceptMappedClaims` on the app registration

Without this, EntraID returns a `400` when the mapped token is exchanged. Patch the application object:

```
PATCH https://graph.microsoft.com/v1.0/applications/{appObjectId}
Body: {"api": {"acceptMappedClaims": true}}
```

### 3e. Managing the policy with `az` commands

All policy operations can be performed using `az rest` after logging in with `az login` (or `az login --allow-no-subscriptions` if your account has no subscription).

**Find the policy ID** (if you don't already have it):

```
az rest --method GET \
--uri "https://graph.microsoft.com/v1.0/servicePrincipals/{servicePrincipalId}/claimsMappingPolicies" \
--query "value[0].id, name:displayName"
```

**View the full policy definition:**

```
az rest --method GET \
--uri "https://graph.microsoft.com/v1.0/policies/claimsMappingPolicies/{policyId}"
```

**Update (replace) the policy definition:**

Build the updated definition as a JSON string — note the inner policy object must be serialised as an escaped string inside the `definition` array:

```
az rest --method PATCH \
--uri "https://graph.microsoft.com/v1.0/policies/claimsMappingPolicies/{policyId}" \
--headers "Content-Type=application/json" \
--body '{
  "definition": [
    {
      "ClaimsMappingPolicy": {
        "Version": 1,
        "IncludeBasicClaimSet": true,
        "ClaimsSchema": [
          {
            "Source": "user",
            "ID": "jobtitle",
            "JwtClaimType": "jobtitle"
          },
          {
            "Source": "user",
            "ID": "department",
            "JwtClaimType": "department"
          },
          {
            "Source": "user",
            "ID": "onpremisesamaccountname",
            "JwtClaimType": "uid"
          },
          {
            "Source": "user",
            "ID": "mailnickname",
            "JwtClaimType": "mailnickname"
          },
          {
            "Source": "user",
            "ID": "mail",
            "JwtClaimType": "email_id"
          },
          {
            "Source": "user",
            "ID": "mailnickname",
            "JwtClaimType": "_user"
          },
          {
            "Value": "<workspace-slug>",
            "JwtClaimType": "portkey_workspace"
          },
          {
            "Value": "<organisations-to-sync-uuid>",
            "JwtClaimType": "portkey_oid"
          }
        ]
      }
    }
  ],
  "displayName": "PortkeyClaimsMappingPolicy"
}'
```

The `PATCH` replaces the entire `definition` — include all claims you want, not just the ones that changed.

**Verify the update was applied** by fetching a fresh token and inspecting it at [jwt.ms](#). Changes take effect on the next token issuance; existing cached tokens are unaffected until they expire.

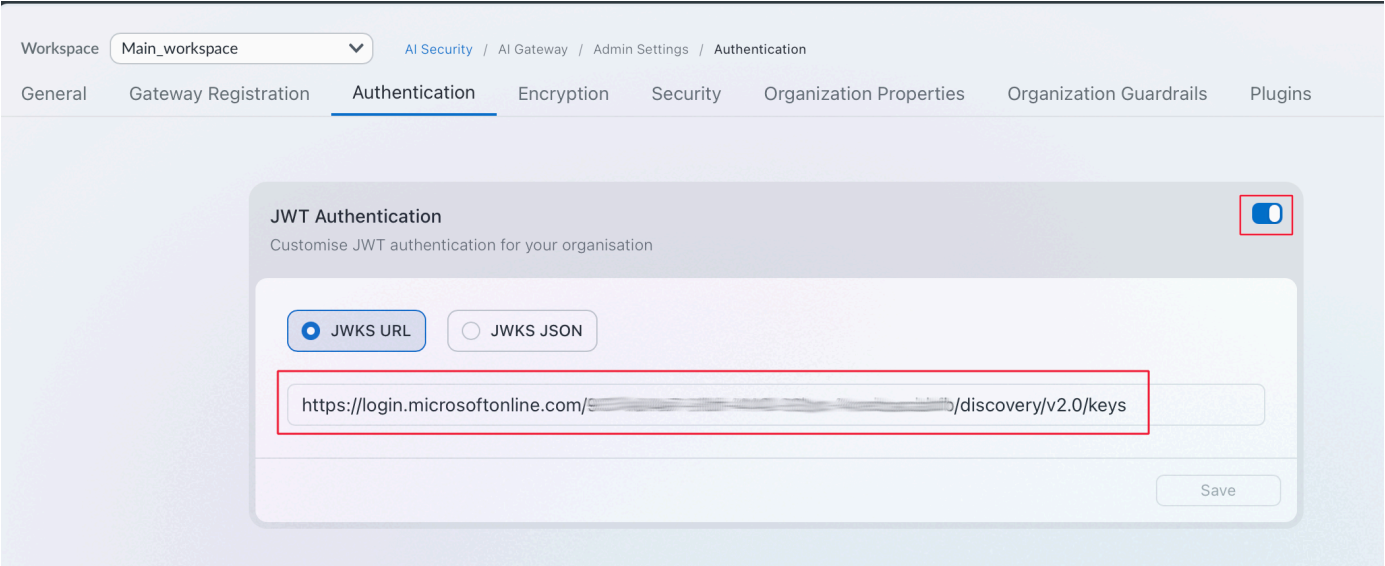
**Shortcut:** Re-running `Setup-EntraID-App.sh` and choosing an existing app will detect the assigned policy, show its current claims, and offer to update it interactively.

4. Configure the Vertex integration

In the AIGW control plane, add an integration for `@vertex` and select the models you want available through this gateway.

5. Enable JWT Authentication

Enable JST Authentication and point it at your EntraID App (replace the tenant ID with your tenant ID). The patching earlier to v2, is due to the URL checking the v2.0 keys.



6. Create a AIGW config

Create a config that routes to the Vertex integration, e.g.:

```
{
  "retry": {
    "attempts": 3
  },
  "cache": {
    "mode": "simple"
  },
  "provider": "@vertex"
}
```

Save it and note the generated **config ID** — it's used in step 6.

7. Configure the 3rd-party (OIDC) provider in Claude Desktop

In Claude Desktop's Developer menu, add a 3rd-party provider pointing at the local hybrid gateway container, with OIDC authentication:

| Field             | Value                                                                |
|-------------------|----------------------------------------------------------------------|
| Base URL          | This gateway's URL (e.g. <code>http://localhost:8787</code> )        |
| Client ID         | The Application (client) ID from step 2                              |
| Issuer URL        | The EntraID tenant issuer URL from step 2                            |
| Bearer token type | ID Token                                                             |
| Scopes            | <code>openid profile email offline_access</code>                     |
| Redirect port     | <code>8080</code> (must match the redirect URI registered in step 2) |

here is a sample configuration:

```
{
  "inferenceGatewayBaseUrl": "http://127.0.0.1:8787",
  "inferenceCustomHeaders": "[redacted]",
  "inferenceGatewayOidcAuthFlow": "browser",
  "inferenceGatewayOidc": {
    "clientId": "9629c204-88db-4e7e-92a8-dda28e451409",
    "issuer": "https://login.microsoftonline.com/<TenantID>/v2.0",
    "bearerTokenType": "id_token",
    "scopes": "openid profile email offline_access",
    "appendOfflineAccess": true,
    "redirectPort": 8080
  },
  "chatTabEnabled": true,
  "modelDiscoveryEnabled": true,
  "inferenceModels": [],
  "inferenceProvider": "gateway",
  "inferenceCredentialKind": "interactive"
}
```

Finally, add a custom header so every request carries the AIGW config from step 5:

```
x-portkey-config: <config-id>
```

## Verifying it works

1. In Claude Desktop, trigger the OIDC login flow for the provider you just added — it should open a browser to EntraID, authenticate, and redirect back to **localhost:8080**.
2. Send a message through Claude Desktop using this provider.
3. Confirm requests are landing in AIGW's control plane logs/analytics for the mapped workspace.

The screenshot shows the 'Observability' section of the AI Gateway control plane, specifically the 'Logs' tab. The breadcrumb navigation indicates the path: AI Security / AI Gateway / Observability / Logs. The page title is 'Observability' and the subtitle is 'AI Gateway Observability'. Below the title are tabs for 'Analytics', 'Logs' (which is selected), and 'Exports'. A search bar with a 'Search Filter' placeholder and a 'Last 24 hours' time range selector are visible. A table of logs is displayed with columns: TIMESTAMP, MODEL, PATH, USER, and TOKENS (COST). One log entry is shown for August 12, 07:18:34 PM, for the model 'claude-sonnet-4-6', with user 'mgollop@sase-au.io' and a cost of 13.8K tokens (8.27 cents).

| TIMESTAMP           | MODEL             | PATH | USER                 | TOKENS (COST)             |
|---------------------|-------------------|------|----------------------|---------------------------|
| Aug 12, 07:18:34 PM | claude-sonnet-4-6 |      | M mgollop@sase-au.io | 13.8K tokens (8.27 cents) |

## Troubleshooting

- **401 Unauthorized** — verify **JWT\_ENABLED: ON** and **JWT\_LOCAL\_AUTH\_DEFAULT\_SCOPES** are set on the gateway (see [docker-compose.yml](#)), and that the ID token actually contains the custom claims from step 3 (decode it at [jwt.ms](#) to check).
- **Org/workspace not resolved** — double check **portkey\_oid** in the token matches this deployment's **ORGANISATIONS\_TO\_SYNC** env var exactly. Decode the token at [jwt.ms](#) to inspect the raw claims.
- **Redirect fails / stuck on EntraID** — confirm the redirect URI registered in EntraID (step 2) exactly matches the redirect port configured in Claude Desktop (step 6), including **http://localhost:<port>**.